

Assignment 01

Course Code:	CSE205 and CSE206
Course Title:	Object-Oriented Programming
Number of Tasks:	8

Vs Code Installation Guide for Java :

<https://youtu.be/VUcI3Y1Nnak>

Task 1

Write a Java program that displays the 2 rightmost digits of your student ID in reverse order. For example, if your student id is 23221454, you need to print 4, and then 5.

Sample Input	Output
23221454	4 5

Task 2

Write the Java code of a program to find the largest among three different numbers entered by the user.

Sample Input	Output
100 23 -4	Largest number: 100
5 17 -5	Largest number: 17

Task 3

Write a Java code that will take a year as input and print whether that year is a leap year or not.

- A year may be a leap year if it is evenly divisible by 4.
- Years that are divisible by 100 (century years such as 1900 or 2100) cannot be leap years unless they are also divisible by 400 (1600 or 2000).

Sample Input	Output
2020	2020 is a leap year
2001	2001 is not a leap year
1900	1900 is not a leap year
2000	2000 is a leap year

Task 4

Write a Java program that takes the CGPA and no of credits completed by a student and prints whether the student is eligible for a waiver and of what percentage.

To be eligible for a waiver, a student must have completed at least 30 credits and earned a CGPA greater or equal to 3.8. If not, please print "The student is not eligible for a waiver".

CGPA	Waiver percentage
3.80 - 3.89	25 percent
3.90 - 3.94	50 percent
3.95 - 3.99	75 percent
4.00	100 percent

Now let's look at the samples.

Sample Input	Output
3.93 78	The student is eligible for a waiver of 50 percent
3.79 24	The student is not eligible for a waiver

Task 5

Write a java program that takes 10 inputs from the user in a loop, and displays the sum, average, minimum and maximum of those numbers.

Sample Input	Sample Output
1 4 2 9 2 -4 3 -1 0 1	Sum = 17 Minimum = -4 Maximum = 9 Average = 1.7
23 2 -4 0 8 12 34 -11 53 21	Sum = 138 Minimum = -11 Maximum = 53 Average = 13.8

Task 6

Write a Java program to print the following patterns

(a) Square pattern

Sample Input	Sample Output
5	***** ***** *****

	***** *****
--	----------------

(b) Triangle pattern

Sample Input	Sample Output
5	* ** *** **** *****

Task 7

Write a Java program that takes a string input in small letters from the user and prints the previous alphabet in sequence for each alphabet found in the input.

Sample Input	Output
wxyz	vwxy
thecow	sgdbnv
abcd	zabc

Task 8

Given a string, create and print a new string with all the consecutive duplicates removed.

Sample Input	Output
ABBCCCCBBAB	ABCBAB
AAABBBBCDDBBECE	ABCDBECE